

BALTIMORE-WASHINGTON, MD, USA

WMO: 724060

Lat: 39.173N

Lon: 76.684W

Elev: 48

StdP: 100.75

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93721

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.3	-8.0	-19.8	0.7	-8.0	-17.1	0.8	-6.3	11.8	2.0	10.7	0.7	3.6	290	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.5	23.9	32.9	23.4	31.5	22.7	25.5	31.6	24.8	30.2	24.2	28.9	4.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.0	27.5	23.4	18.3	26.9	22.8	17.6	26.5	78.3	31.8	75.5	30.2	72.9	29.0	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.4	7.4	-13.9	36.7	2.5	2.0	-15.7	38.1	-17.2	39.3	-18.6	40.4	-20.5	41.8		
(5)			-14.8	26.8	2.4	1.0	-16.5	27.5	-17.9	28.1	-19.3	28.6	-21.0	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	1.1	2.5	6.9	13.0	18.0	23.1	25.5	24.5	20.8	14.3	8.3	3.8	
	DBStd	9.57	5.63	5.17	5.22	4.72	4.31	3.46	2.79	2.72	3.63	4.46	4.56	4.79	
	HDD10.0	943	281	216	127	21	1	0	0	0	0	11	86	203	
	HDD18.3	2486	534	443	357	173	62	5	0	0	17	142	302	452	
	CDD10.0	2228	6	6	29	109	248	392	481	449	323	143	34	9	
	CDD18.3	730	0	0	1	12	51	147	223	191	90	16	1	0	
	CDH23.3	6759	0	1	18	139	477	1367	2243	1714	681	115	4	0	
	CDH26.7	2566	0	0	3	39	153	520	972	647	208	24	0	0	
Wind	WSAvg	3.0	3.3	3.4	3.7	3.6	3.0	2.7	2.5	2.3	2.5	2.6	2.9	3.0	
Precipitation	PrecAvg	1088	77	77	94	82	98	97	105	99	102	86	81	90	
	PrecMax	1823	199	182	220	218	221	333	425	277	338	234	207	205	
	PrecMin	717	12	7	5	10	9	21	18	23	5	0	8	16	
	PrecStd	220	38	42	44	39	50	58	65	55	74	55	47	47	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	21.1	26.0	30.6	32.6	35.4	36.9	35.7	33.5	29.3	23.4	20.4	
		MCWB	14.5	14.3	16.1	19.2	22.2	23.8	25.0	24.5	22.6	21.4	17.0	16.2	
	2%	DB	15.9	17.0	21.7	27.4	30.4	33.2	34.9	33.5	31.2	26.5	20.5	16.8	
		MCWB	12.9	11.3	14.3	17.1	21.3	23.3	24.3	23.9	22.1	19.8	15.3	13.8	
	5%	DB	12.6	13.4	18.6	24.2	28.3	31.5	33.2	31.8	29.1	23.9	18.2	13.9	
		MCWB	9.4	8.7	12.4	16.1	20.3	22.6	23.8	23.2	21.3	18.5	14.0	10.8	
	10%	DB	9.3	10.4	15.7	21.6	26.0	29.7	31.6	30.2	27.1	21.9	16.0	11.1	
		MCWB	6.3	6.7	10.4	14.8	19.1	22.0	23.2	22.4	20.4	17.8	12.3	8.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.5	17.5	20.8	23.8	25.9	26.6	26.2	25.0	23.4	19.0	17.5
			MCDB	17.5	20.2	22.7	26.1	30.4	31.5	33.6	32.3	29.5	26.1	21.2	19.4
2%		WB	13.5	12.6	15.5	19.1	22.5	24.7	25.7	25.3	24.0	21.6	16.9	14.6	
		MCDB	15.6	15.8	20.0	24.6	28.4	30.2	32.2	31.1	27.8	24.8	19.3	16.5	
5%		WB	10.0	9.4	13.5	17.5	21.4	24.0	25.0	24.6	23.2	20.2	15.0	11.5	
		MCDB	12.2	12.4	17.6	22.6	26.6	29.1	30.8	29.4	26.6	23.1	17.4	13.2	
10%		WB	6.8	7.0	11.1	15.8	20.2	23.2	24.4	23.9	22.3	18.1	13.0	8.7	
		MCDB	8.8	10.2	14.8	20.7	24.5	27.9	29.4	27.9	25.5	21.0	15.2	10.5	
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	10.4	11.5	10.9	10.6	10.3	10.1	10.2	10.7	10.2	8.7	
		MCDBR	12.7	14.3	14.9	15.5	13.7	12.6	12.1	11.8	12.6	13.0	13.3	12.2	
		MCWBR	9.9	10.0	9.3	7.8	6.4	4.9	4.1	4.3	5.1	7.0	8.7	9.6	
	5% WB	MCDBR	11.9	12.7	13.3	12.9	12.1	10.6	10.3	10.0	9.4	10.2	10.5	10.6	
		MCWBR	10.2	10.0	9.4	7.5	6.1	4.8	4.1	4.2	4.6	6.5	8.4	9.6	
Clear-Sky Solar Irradiance	taub	0.309	0.320	0.357	0.398	0.441	0.494	0.514	0.509	0.426	0.361	0.324	0.309		
	taud	2.477	2.430	2.391	2.298	2.207	2.099	2.062	2.061	2.266	2.451	2.515	2.514		
	Ebn at Noon	866	904	899	879	844	796	776	768	816	844	842	839		
	Edh at Noon	79	94	109	128	143	159	164	160	123	92	76	71		
All-Sky Solar Radiation	RadAvg	2.04	2.89	3.84	4.91	5.51	6.11	6.03	5.35	4.36	3.23	2.33	1.79		
	RadStd	0.17	0.33	0.28	0.39	0.65	0.47	0.30	0.30	0.51	0.35	0.20	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+72
(47)	Regional (79 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+78

Nomenclature: See separate page